Paper:

Three-DOF Electrostatic Induction Actuator Providing Translational and Rotational Surface-Drive Motion

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This paper describes the driving characteristics of a three degree-of-freedom (three-DOF) electrostatic induction actuator, which can demonstrate surface-drive characteristics with translational and rotational motions. It consists of a sheet-type slider without electrodes and a planar stator with an array of three-phase driving electrodes. The electrodes with different orientations are aligned in a regular manner to construct a four-by-four checkerboard pattern. Controlling applied voltage patterns can generate translational or rotational patterns of electrostatic fields, which drive the slider. The performance of the three-DOF actuator with regards to translational and rotational motion was investigated.

Keywords: electrostatic actuator, induction actuator, Three-DOF, surface-drive

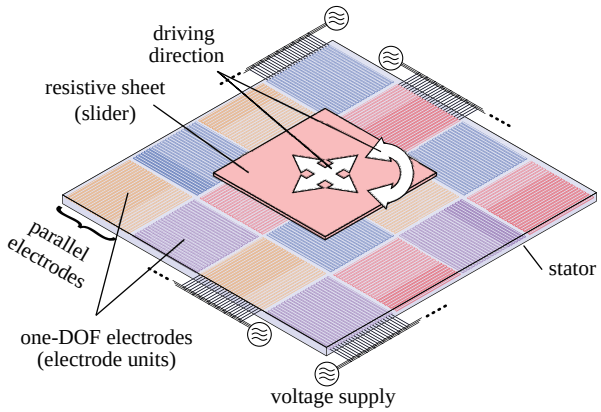


Fig. 1. Basic structure of the three-DOF electrostatic actuator. The four-by-four tiled pattern of one-DOF electrodes is shown.

1. Introduction

Advanced features of actuators have the possibility of improving product designs and structures. This is because the performance of dynamic products largely depends on the features of their actuators. Therefore, various actuators have been studied for wide applications such as robotics [1, 2], power assist [3, 4] and human-computer interaction [5–7]. In such various fields, multiple-degree-of-freedom (multi-DOF) actuators [8–10] and flat-type actuators [11–13] are components that have substantial potential for producing new designs of products. Although electromagnetic actuators provide a high level of performance, have a wide variation of their design, and are highly reliable, their bulky and heavy structures sometimes limit or nullify the requirements of a product design.

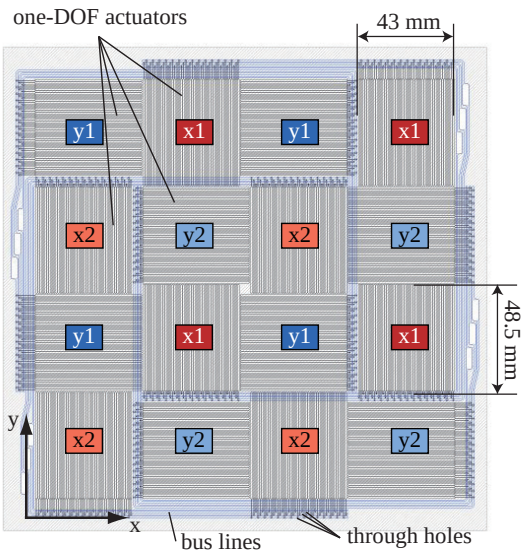
Electrostatic film actuators are one candidate for filling such requirements due to their thin, flat, and lightweight structures [15–18]. Several types of electrostatic film actuators have been studied, including types that have exhibited high output [16] or untethered features [15, 17]. The specific features have shown possibilities for tangible applications [5–7, 18].

Two-DOF translation and rotation using an electrostatic film actuator has been achieved by combining four electrode panels [19]. However, the structure was able to generate only one point of rotation at the center. In addition, its *xy*-translation was accompanied by undesirable rotation of the slider due to imbalance of forces from the electrodes. These limitations were improved by increasing the number of the panels [7]. A prototype four-by-four electrode panel was fabricated, which demonstrated *xy*-translation as well as rotation. However, the dynamic features were not investigated in detail.

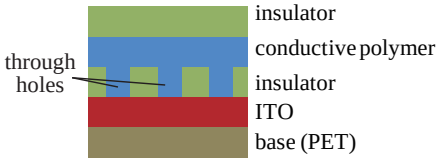
In this study, we investigate the motion features of the three-DOF actuator designed in the study [7] in detail. The appropriate size of the slider, the variation of the motion with time, and rotation at multiple points of rotation are investigated.

2. Basic Structure and Driving Principle of Electrostatic Actuator

The three-DOF actuator described in this paper comprised an arrayed structure of one-DOF actuators (electrode units). **Fig. 1** shows a schematic diagram of the basic structure. It consists of two components, the stator and slider. The stator comprises embedded electrodes whereas the slider does not have electrodes. The stator electrodes have a three-phase structure with parallel electrodes, allowing supply of three-phase AC voltage. The slider is made of a highly resistive material, such as plastic or paper sheet.



(a) top view of stator electrodes



(b) cross-section of a stator electrode

Fig. 2. Structure of a three-DOF electrode designed using a combination of one-DOF electrodes.

Three-phase voltage applied to the stator electrodes generates a traveling voltage wave on the stator. In addition, it induces a traveling electric charge wave on the slider. Interaction between these two waves generates horizontal thrust force on the slider. Changing the voltage frequency and phase can alter the speed and direction of the force. Since the interaction also generates an attractive force between the stator and the slider, friction between them should be kept to a minimum.

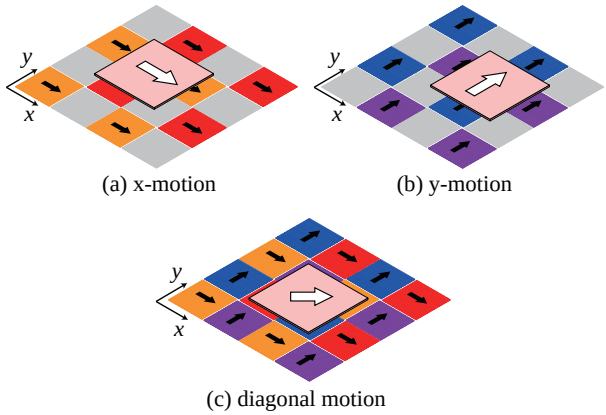
3. Three-DOF Actuator

3.1. Structure

The design of the three-DOF actuator comprised tiled one-DOF actuators in a checkerboard pattern. **Fig. 2** shows the structure, comprising four-by-four one-DOF actuators with electric bus lines. Half of the actuators were oriented in the *x*-direction, whereas the other half were oriented in the *y*-direction. The actuators with the same indices were connected to the same bus lines, hence they were applied the same AC voltage patterns. Voltage patterns to the four groups of actuators were controlled independently to generate *xy*-translation and rotation of the slider.

3.2. Driving Methods

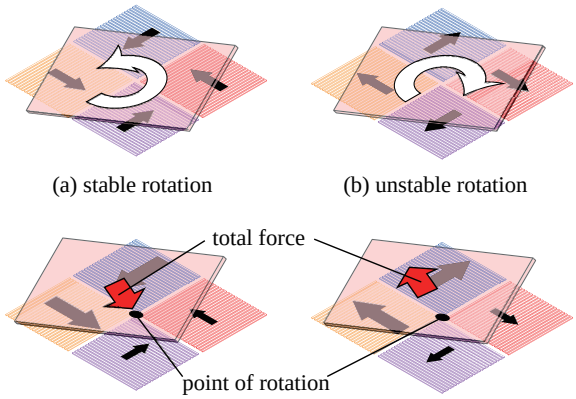
Figure 3 shows the method for generating translational motion. For *x*- or *y*-direction driving, the electrodes producing a corresponding direction of electrostatic force were activated. For diagonal motion, electrodes producing both directions of electrostatic force were used simultaneously. Exchanging any two phases of the three-phase voltage could reverse the direction of voltage waves, which resulted in the slider changing direction.



(a) x-motion (b) y-motion

(c) diagonal motion

Fig. 3. Driving methods for *x*-, *y*-, and diagonal translation.



(a) stable rotation (b) unstable rotation

(c) deviation during stable rotation (d) deviation during unstable rotation

Fig. 4. Driving method for rotation and resulting stability. (a) Stable and (b) unstable rotation around a point of rotation. (c) Restoring force during stable rotation. (d) Outward force during unstable rotation.

Rotational motion could be achieved by using a combination of four electrodes as shown in **Fig. 4**. A combination of opposing forces in the *x*-direction and opposing forces in the *y*-direction could generate rotation around the center of the four electrodes. Thus, rotation could be achieved at limited points. In addition, only one direction of rotation at each of these points was stable. This is because, during unstable rotation, deviation of the slider enhances the outward force from the point of rotation. The resulting force on the slider for stable and unstable rotation are compared in **Figs. 4(c)** and **(d)**, respectively. The stable directions of rotation and their force patterns for the entire three-DOF actuator are summarized in **Fig. 5**. Four different categories of stable points of rotation could be produced on the stator by changing the applied volt-

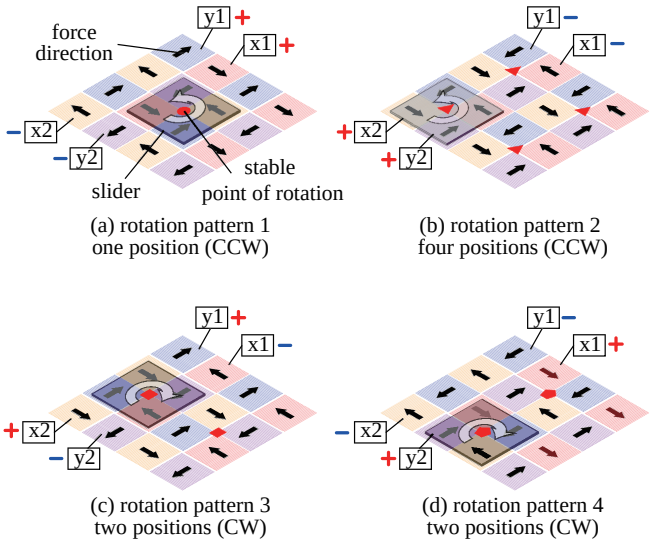


Fig. 5. Available stable counter-clockwise (CCW) or clockwise (CW) rotation points for a three-DOF actuator. Four categories of points of rotation could be produced on the stator by changing applied voltage patterns. Two of them produce stable CCW rotation, and the other two produce stable CW rotation.

age patterns. Two categories produced stable counter-clockwise (CCW) rotation, and the other two produced stable clockwise (CW) rotation.

3.3. Fabrication of Stator

The fabrication method of the three-DOF actuator, shown in **Fig. 6** is described in this section. Electrodes other than the bus lines were fabricated on a polyethylene terephthalate (PET) sheet using an etching of indium tin oxide (ITO) electrodes. The electrodes other than at the through-hole positions were then covered with an insulating ink using the screen printing process. Bus lines were printed on the insulating layer and through-holes so as to connect them to the corresponding phases of the ITO electrodes. Conductive polymer ink was used for the bus lines. After this process, the entire area of the actuator other than the external connection terminals was re-covered with insulating ink.

The pitch of the driving electrodes was 1.0 mm, comprising a line/space width ratio of 0.7 mm / 0.3 mm. The size of each one-DOF electrode measured 48.5 mm by 43 mm.

4. Experimental Evaluation of Actuator

In this section, the motion characteristics of the three-DOF actuator are evaluated experimentally. The dynamic motion of the sliders was measured using optical tracking.

For the sliders, transparent plastic sheet, used for over-head projectors (OH-001, Plus Corp., Base material: Polyester, thickness: 0.1 mm), was cut into eleven different-sized squares. The sides of the squares ranged

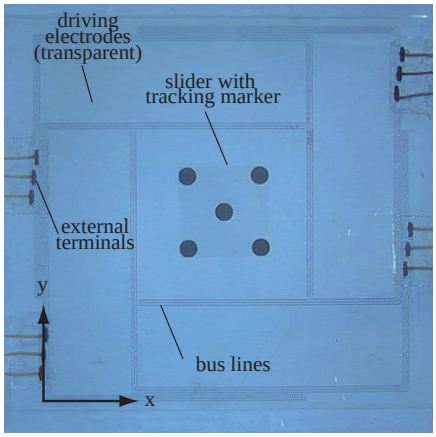


Fig. 6. Top view of a fabricated three-DOF actuator.

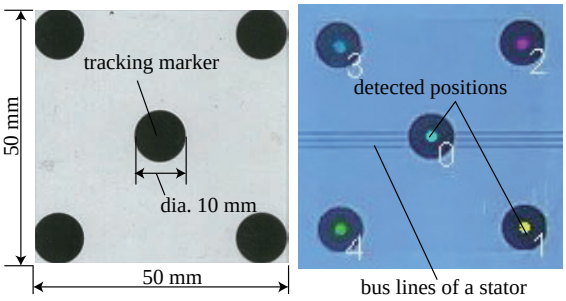


Fig. 7. Slider, constructed out of plastic sheet, with sides of 50 mm. The center and each of the four corners has a tracking marker for motion tracking.

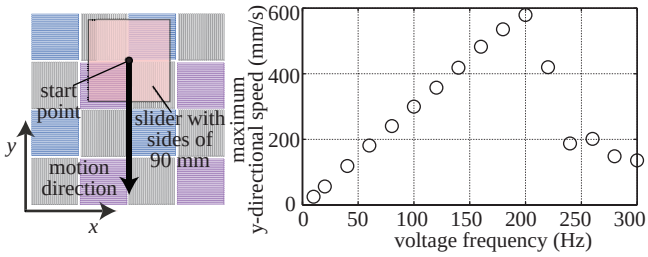
from 20 mm to 120 mm in 10 mm increments. The surface resistance, which has been found to be the most important parameter for the dynamic performance [19, 20], was $5.3 \times 10^{11}\,\Omega$. A slider with 50 mm sides is shown in **Fig. 7**. Five tracking markers were placed on each slider: one at each corner and one at the center. The diameter of the markers was 10 mm, other than the markers on the slider with sides of 20 mm, which had a diameter of 5 mm.

Slider motion was captured using a Motion Analyzing Microscope (VM-6000, Keyence Corp.). High frame rate video with a frame rate of 250 fps was used, and the marker positions were then tracked. Driving voltages were generated using a waveform generator (7075, Hioki E. E. Corp.) and amplified a thousand-fold using high-voltage amplifiers (Model 609C-6, TReK Inc.). The voltages consisted of three-phase sinusoidal waves with an amplitude of 600 V_{0-p} . Glass beads with a diameter of 100 μm were scattered between the stator and the sliders to reduce friction.

4.1. X and Y-Translation

4.1.1. Voltage Frequency Conditions

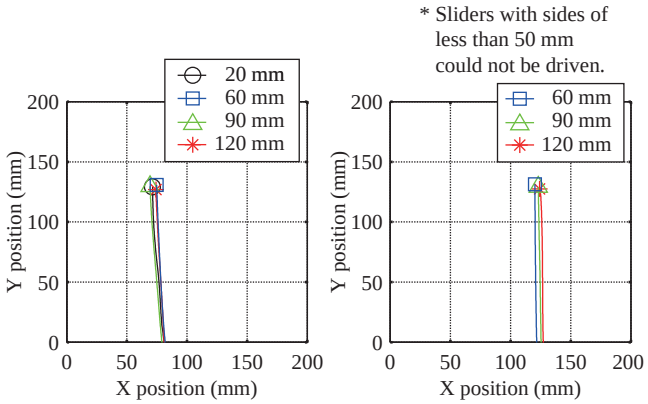
First, we investigated the applicable voltage frequency for driving the slider. A slider with sides of 90 mm was placed at the center of four electrodes, as shown in



(a) y-motion and the start point

(b) maximum y-directional speed against voltage frequency

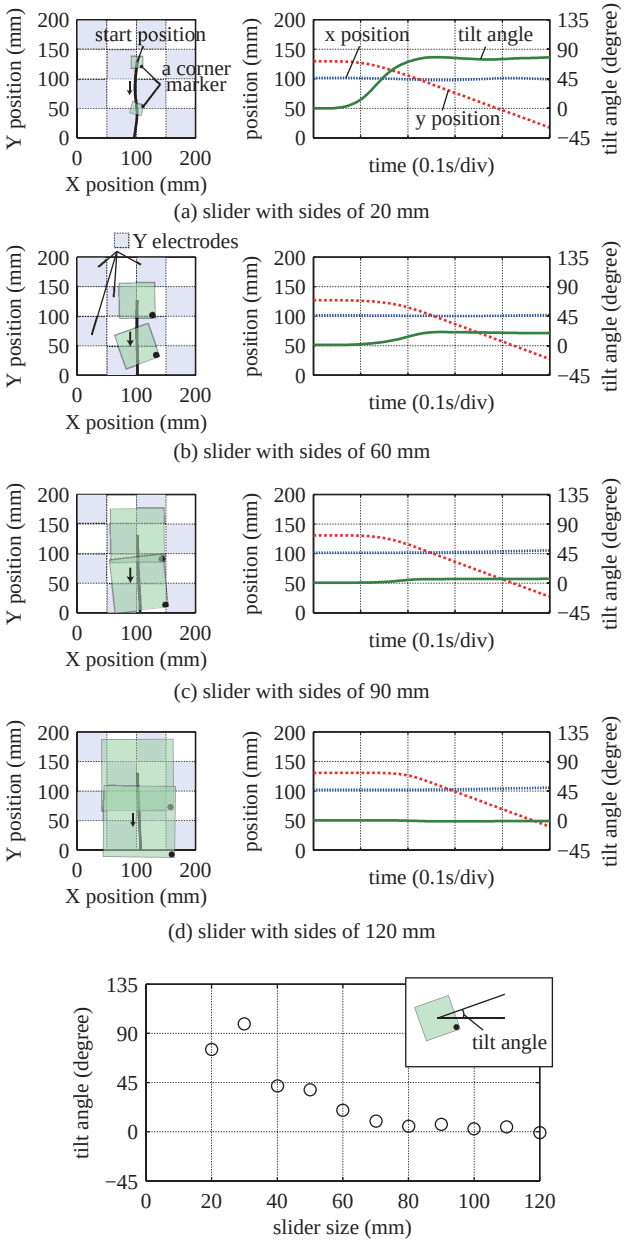
Fig. 8. Y-translation speed for different voltage frequencies for a slider with sides of 90 mm.



(a) from y-directional electrode

(b) from x-directional electrode

Fig. 9. Y-translation trajectories of different-sized sliders from *x*- and *y*-directional electrodes. Trajectories from each electrode are summarized.



(a) slider with sides of 20 mm

(b) slider with sides of 60 mm

(c) slider with sides of 90 mm

(d) slider with sides of 120 mm

(e) average of tilt angle at position between y=50 and y=75

Fig. 10. Tilting of sliders during *y*-translation, starting from an edge between an *x*- and a *y*-directional electrode.

Fig. 8(a). Different frequencies of voltage from 10 to 300 Hz were then applied. The resultant maximum *y*-directional speeds are shown in **Fig. 8(b)**. The maximum speed showed one peak as well as a proportional trend in the lower frequency region. Although the tendency was similar to the results of a previous study [21], in this study the actuator exhibited quasi-synchronous speeds of $3pf$, in which p and f represent the electrode pitch and voltage frequency, respectively. The difference of characteristics compared with the previous study are caused by the difference of stator pitch and the surface resistance.

For the following experiments, a voltage frequency of 100 Hz was used.

4.1.2. Slider Sizes for X and Y-Translation

Next, the possible slider sizes allowing translational motion were investigated. Since each electrode unit was only able to drive one-DOF motion, a slider of insufficient size placed on a different directional electrode was not able to be driven. In contrast, a slider that was larger than each unit was able to be driven because some parts of the slider were always on appropriate directional electrodes.

The eleven different-sized sliders were placed at the center of an *x*- or *y*-directional electrode, and it was attempted to drive the slider in the *y*-direction. **Fig. 9** shows the resulting trajectories of translational motion of the sliders with sides of 20, 60, 90, and 120 mm. Note that the origin of the *xy*-coordinate system is at the bottom left corner of the stator, as previously shown in **Fig. 6**. From the *y*-directional electrode, driving was possible for all sizes of slider. However, from the *x*-directional electrode, small-sized sliders with sides of less than 50 mm could not be driven, whereas those with sides of 60 mm or more could be successfully driven from the position. Since parts of the larger sliders were overlapping onto *y*-directional electrodes, they successfully underwent *y*-motion.

The *y*-motion commencing from an edge between an *x*- and a *y*-electrode was also examined, and is shown in **Fig. 10**. **Figs. 10(a)-(d)** shows the motion undergone

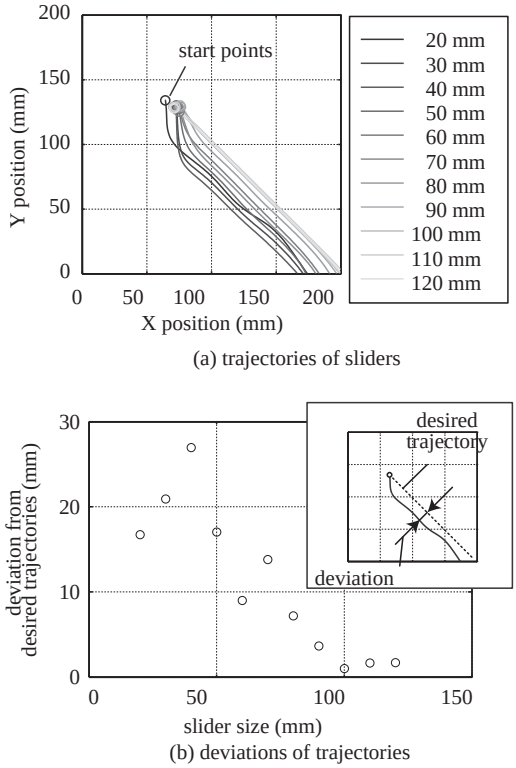


Fig. 11. Diagonal translation of different-sized sliders from *y*-directional electrodes.

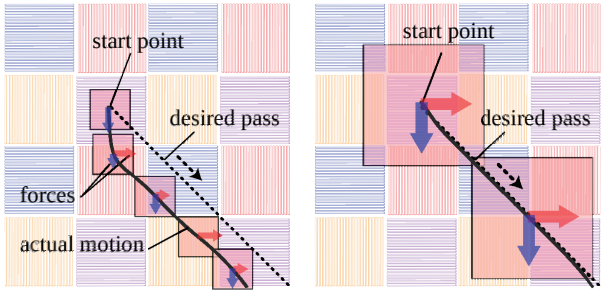


Fig. 12. Comparison of diagonal motion between a small-sized slider (a) and a large-sized slider (b).

by the sliders with sides of 20, 60, 90, and 120 mm, respectively. Due to the existence of static friction and the imbalance of *y*-directional forces, the smaller-sized sliders tilted near the start point. However, the larger-sized sliders exhibited smaller tilting, since they were overlapping onto other *y*-electrodes which cancelled their rotational moment. **Fig. 10(e)** shows a graph of the relationship between tilt angle and slider size. It can be seen that sliders with sizes closely approximating the size of four electrode units experienced sufficiently decreased tilting motion owing to the balanced moment.

The diagonal motion of different-sized sliders was also investigated. A slider was placed at the center of a *y*-directional electrode, and it was subsequently driven in a direction towards the bottom right. The trajectories of all sizes of sliders are shown together in **Fig. 11(a)**. Although the intended motion was diagonal, small-sized sliders initially moved in the *y*-direction because no part of their area was overlapping with *x*-direction electrodes, as shown in **Fig. 12(a)**. They underwent diagonal motion after arriving at the boundary between two electrodes. Larger sliders with sides greater than 90 mm underwent diagonal motion without showing the initial *y*-direction motion, because they were constantly subjected to forces in both directions, as shown in **Fig. 12(b)**. **Fig. 11(b)** shows the deviation from the desired trajectory according to slider size. As shown in **Fig. 11(b)**, the slider with sides of 40 mm showed the largest deviation. This was probably caused by the following two reasons: it could accelerate only in the *y*-direction at the start, whereas the 50 mm slider could not, and its relatively large size compared with the 20 or 30 mm sliders enabled it to receive *y*-directional force for a longer period after crossing into the next *x*-directional electrode.

These results showed that the three-DOF actuator could achieve the desired translation of the slider if the size of the slider was sufficiently large compared to the electrode unit.

4.1.3. Bidirectional Translation

Bidirectional *x*- and *y*-motion was investigated, and the results are shown in **Figs. 13** and **14**, respectively. A slider with sides of 90 mm was used, and the center of the slider was tracked. The directions were changed by altering voltage conditions for the four electrode groups. The trajectory of the center of the slider, as well as the variation of the *x*- and *y*-positions with time, are shown in each figure. The circle in each figure represents the start point of the center of the slider. As shown in the figures, the actuator successfully achieved bidirectional *x*- and *y*-motion.

Multi-directional diagonal motion was also investigated, and the results are shown in **Fig. 15**. The slider and the setup were the same as that described above. A combination of voltage patterns applied to the *x*- and *y*-electrodes realized diagonal motion in four directions. The results showed that sliders of appropriate sizes were able to be driven in any diagonal direction using the methods shown in **Fig. 3**.

4.1.4. Initial Positions for X- and Y-Translation

Since actuators require relatively large thrust at their start-up, the start points of the slider are important for generating sufficient thrust. In this section, the influence of different start points is investigated for a slider with sides of 90 mm. **Figs. 16(a)**, **(b)**, and **(c)** show the influence of different start points on the trajectories of *x*-, *y*-, and diagonal motion, respectively. The curved motion experienced for the *y*-translation may have been caused by a slight slope around the electric terminals due to a protrusion of insulating material. These results show that

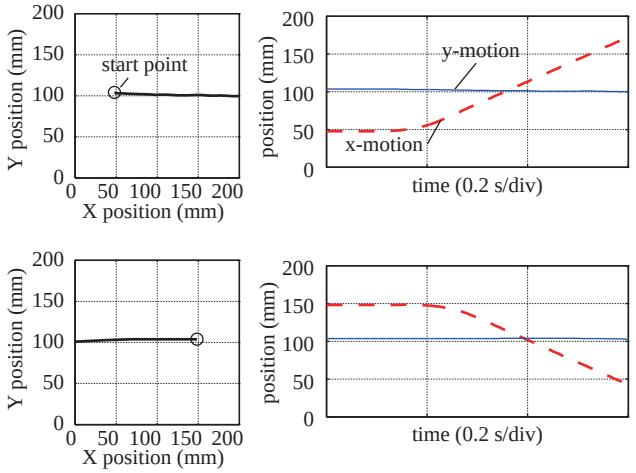


Fig. 13. Bidirectional *x*-translation for a slider with sides of 90 mm.

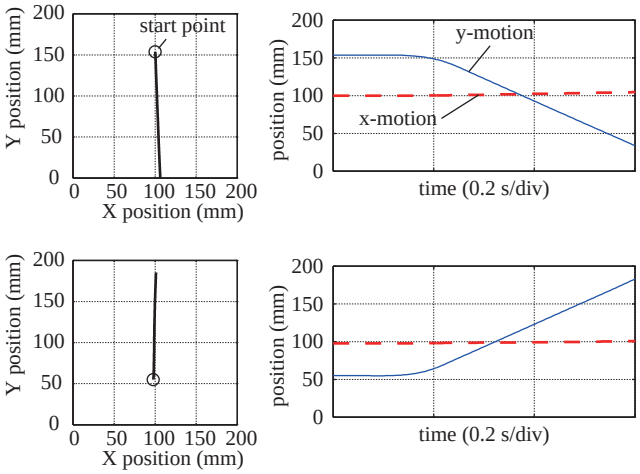


Fig. 14. Bidirectional *y*-translation for a slider with sides of 90 mm.

translational motion can be affected from any start position without showing an initial deviation in the *x*- or *y*-directions.

4.2. Rotation

Rotational motion was evaluated using the sliders with sides of 20, 50, and 90 mm.

4.2.1. Stability of Rotation

First, the stability of rotation was investigated at the point of rotation. **Fig. 17** shows trajectories of the center and a corner of the slider with sides of 90 mm. It underwent stable rotational motion, shown in **Fig. 17(a)**, and unstable rotational motion, shown in **Fig. 17(b)**, at a point of rotation at which CCW motion was predicted as the stable direction of rotation (see **Fig. 5**).

4.2.2. Stable Rotation at Various Points of Rotation

Figures 18(a)-(d) show rotation of the slider with sides of 50 mm at all nine different points of rotation. The slider

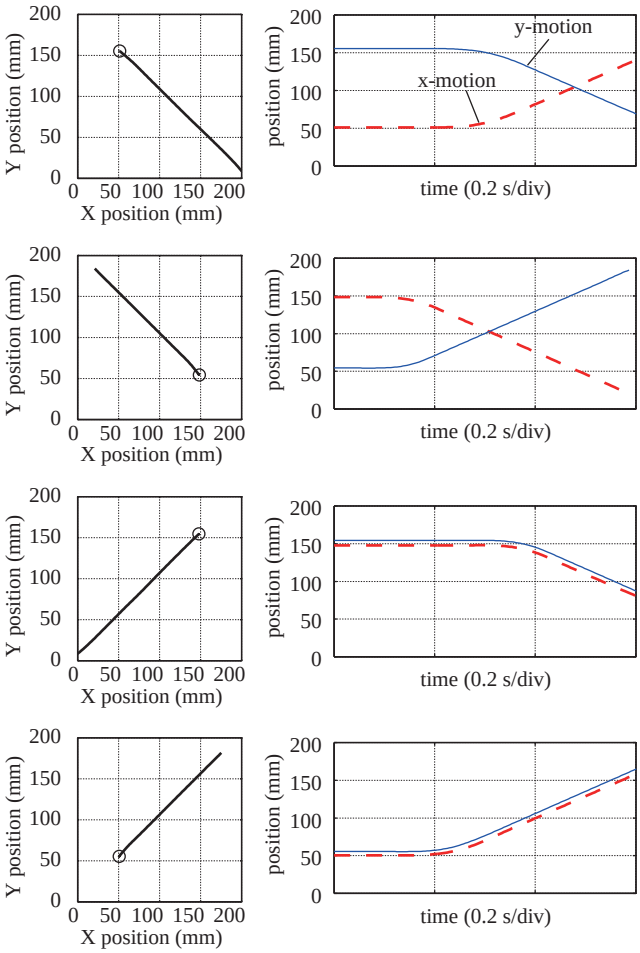


Fig. 15. Four-directional diagonal translation for a slider with sides of 90 mm.

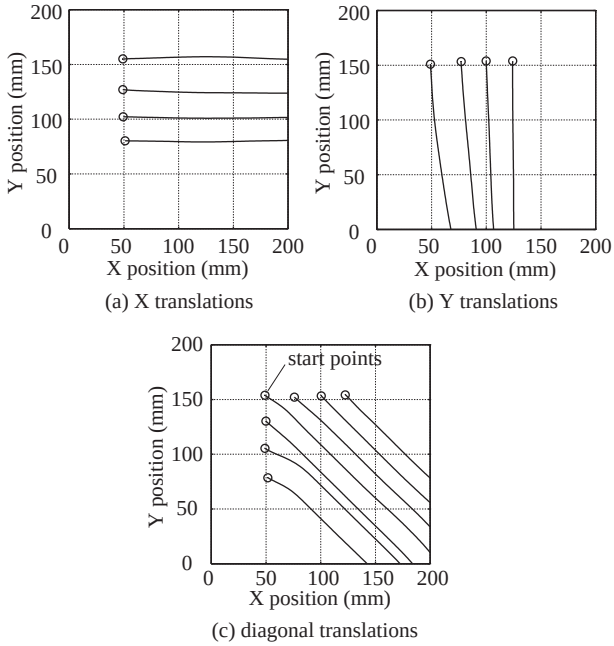


Fig. 16. Trajectories of *x*-, *y*-, and diagonal translation from different start points for a slider with sides of 90 mm.

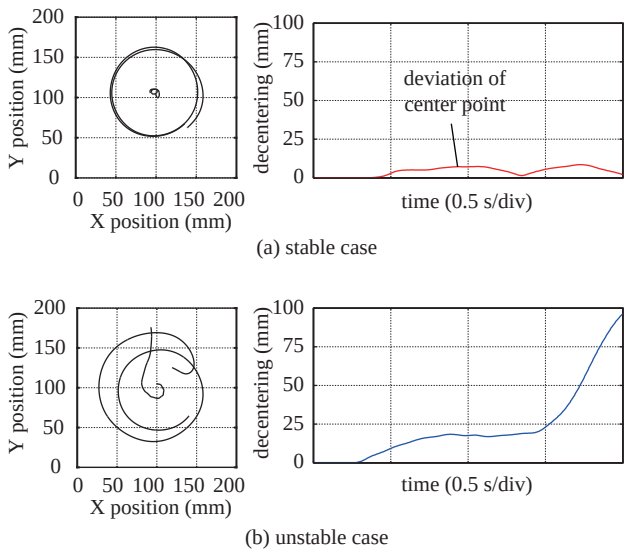


Fig. 17. Stability of rotation. Stable rotation (a) and unstable rotation (b) are shown for the slider with sides of 90 mm. The trajectories of the center and a corner of the slider are shown.

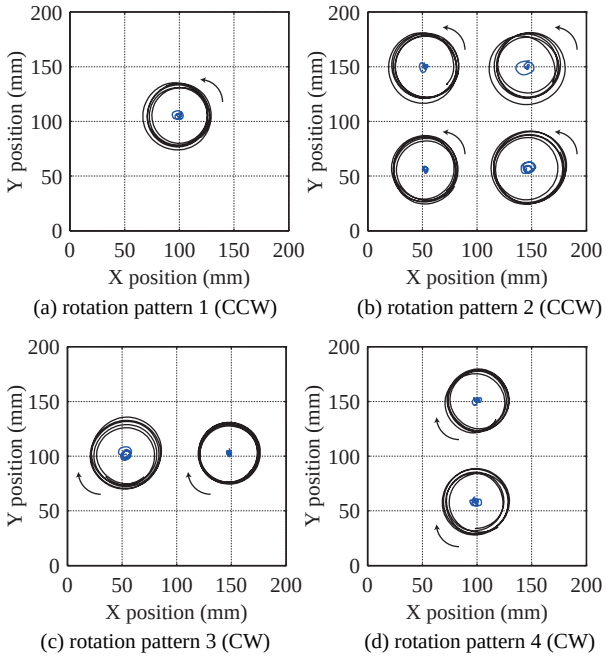


Fig. 18. Stable rotation of slider with sides of 50 mm at all nine points of rotation. Directions of rotation according to the directions shown in **Fig. 5** were demonstrated.

underwent CW or CCW rotation at each point, according to the directions shown in **Fig. 5**. Although the points of rotation on the stator were limited to nine points of rotation and the directions of rotation were limited to one direction at each point of rotation, rotational motion was clearly demonstrated.

4.2.3. Effect of Slider Size on Rotational Stability

The rotation frequency of sliders with sides of 20, 50, and 90 mm are shown in **Fig. 19(a)**. The point of rotation

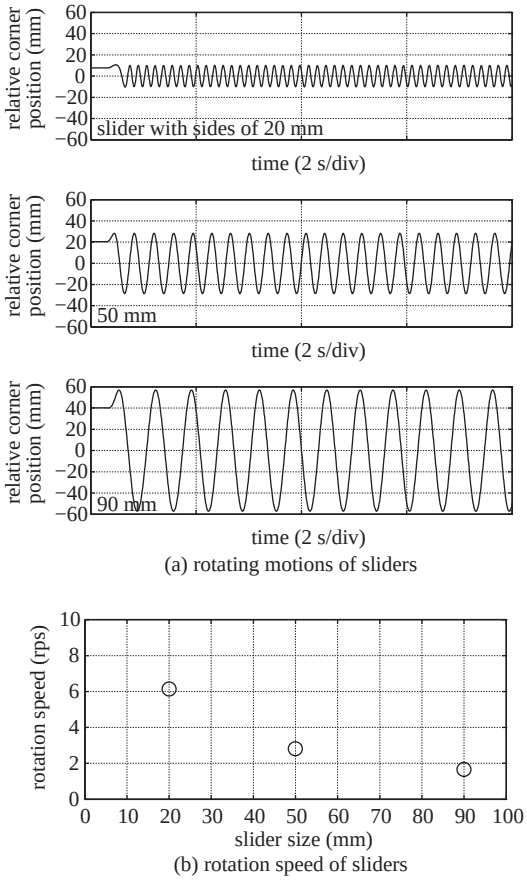


Fig. 19. Rotation of different-sized sliders. Frequency of relative corner position for different sized sliders (a) and comparison of rotation speed with size of slider (b) are shown.

was the center of the stator. The relative x position of a corner marker from the center is plotted against time. As seen in the graphs, rotational speeds became stable within a second for all sizes of slider. **Fig. 19(b)** shows a comparison of rotational speeds with size of slider. Although larger-sized sliders were able to increase their torque, the moment of inertia increased as biquadratic to their size. As a result, the rotation speeds showed a gradual decrease as the size of the slider increased.

5. Conclusions

This study examined the dynamic characteristics of a three-DOF electrostatic induction actuator. A tiled structure of electrodes was proposed, and translation and rotational motion for the three-DOF actuator were investigated. By comparing results obtained for different-sized sliders, it was found that sliders of a sufficiently large size compared to the electrode units exhibited suppressed tilting and deviation from desired directions during translational motion. In addition, it was shown that the sliders were able to demonstrate x-, y-, and diagonal motion from any start position on the stator. Combinations of x- and y-electrodes generated multiple points of rotation, at which

a slider showed stable rotation in the directions predicted. Therefore, we affirm that the newly designed three-DOF actuator successfully demonstrated translation and rotation, although some motion was limited. The limitations are projected to be solved by using finer checkerboard patterns of electrodes.

Acknowledgements

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